RAY KWAN

SENIOR SOFTWARE ENGINEER

Markham, ON ▪ (647) 210 3443 ▪ lokto.kwan@gmail.com

SUMMARY

Senior Software Engineer with 7+ years of experience building precise manufacturing systems and full-stack software. Proven record of leading teams to deliver high-performance CNC and laser vision solutions adopted by numerous NHL teams

WORK EXPERIENCE

LEAD TECHNICAL CONSULTANT
Jan 2026 - present

BAUER HOCKEY

- Supported the development of **Bauer's next-generation** skate tuning machine
- Evaluated and selected software solutions for machine vision, ensuring seamless integration of Skatescribe software with new hardware
- Migrated cloud infrastructure from **Linode** to **AWS** to support scaling from 20+ to 700+ deployed machines globally

LEAD SOFTWARE ENGINEER
June 2023 - present

SKATESCRIBE

- Managed a team of **7 engineers** to develop advanced features on our skate tuning machine, driving adoption among **7+ NHL/AHL teams** and **12+ pro shops**
- Architected to **microservices** refactor to decouple core business logic, decreasing cycle time by **25%** and improving data accuracy by **70%**.

FULL STACK DEVELOPER
May 2021 - June 2023

SKATESCRIBE

- Prototyped skate profiler and sharpener by integrating a laser profiler with a CNC using **Python, React, and PSQL.**
- Invented proprietary algorithms achieving **±1 µm precision**, and built a scalable cloud infrastructure on **Linode** with **Docker** and **NGINX**.

FULL STACK DEVELOPER
Sept 2020 - May 2021

COGNITIVE CENTIVIZER

- Developed **React** and **Node.js** games to measure reaction time and memory, integrating real-time data collection and analytics for 100+ participants.

FULL STACK DEVELOPER
May 2019 - June 2020

AUTOCASE (Internship)

- Developed a cost-benefit analysis platform (**AngularJS** and **Python**) transforming economic research models into interactive visualizations of building design impacts.

SKILLS

Languages & Frameworks
- Python
- Flask
- Javascript
- ReactJS/ NextJS
- AngularJS
- NodeJS
- Java
- PSQL

Libraries
- Numpy/ Scipy
- Pandas
- MUI

DevOps
- Docker
- Linode
- Github Actions

WEBSITES

- Website
- LinkedIn

Projects

- Music Player
- Wordle Copycat
- Graphing Library
- ATM

EDUCATION

B.Sc. Computer Science
University of Toronto
2016 - 2021
Dean's Honor

Digital Growth Marketing
Talent Accelerator